

Binary · AS · Answers

24 written + 6 pen-first = 30 · Student copy

One mark per stated point unless grouped. Accept equivalent correct working. Do not credit contradictory working; TV = too vague, NE = not enough. Width marks apply only where stated below.

1. Denary to binary [4]

(a) $91 = 64 + 16 + 8 + 2 + 1$ [1] → **01011011** [1]. (b) $172 = 128 + 32 + 8 + 4$ [1] → **10101100** [1]. Accept a correct repeated-division method for each working point.

2. Binary to denary [4]

(a) $128 + 32 + 16 + 2 + 1$ [1] = **179** [1]. (b) $64 + 32 + 16 + 8 + 4 + 2$ [1] = **126** [1].

3. Capacity [4]

(a) **256** patterns [1]; largest **255** [1]. (b) **4096** patterns [1]; largest **4095** [1]. Accept equivalent powers for pattern counts.

4. Addition [4]

(a) Correct carry method: carry begins at the 2 column, then propagates through every column to the new 256 column [1]; full sum **100000000** ($154 + 102 = 256$) [1]. (b) **00000000** [1]. (c) 256 exceeds the unsigned eight-bit maximum 255 / needs nine bits [1]. “It overflows” alone is NE.

5. Authored state-register rehearsal [1]

1024 or 2^{10} [1]. Ten independent binary positions give 1024 patterns. Do not confuse the count with the largest unsigned value 1023.

6. Score [4]

(a) $250 + 10 = 260$; positional working $260 = 256 + 4$ (or valid binary addition) [1]; binary **100000100** with denary **260** [1]. (b) **00000100 = 4**, both required [1]. (c) The stated model discards the high bit worth 256, retaining the low eight bits [1]. This is not a claim that every system silently wraps.

7. Representation [3]

(a) Two binary states map to the switch's on/off states [1]. (b) **4** [1]. (c) **8** [1].

8. Pen-first card task [6]

(a) YES, YES, NO, YES, YES, NO; $27 = 16 + 8 + 2 + 1$. Correct sum of weights [1]; all six YES/NO answers correct [1]. (b) **101010** [1]; $32 + 8 + 2 = 42$ [1]. (c) New weight **64** [1]; maximum **127** [1]. Zero is the all-NO pattern.

Extension (unmarked): $022_3 = 2 \times 3 + 2 = 8$; the next count is $100_3 = 9$. Each position carries when it reaches three because its allowed digits are only 0, 1, 2. Karel companion: outside the mark total.